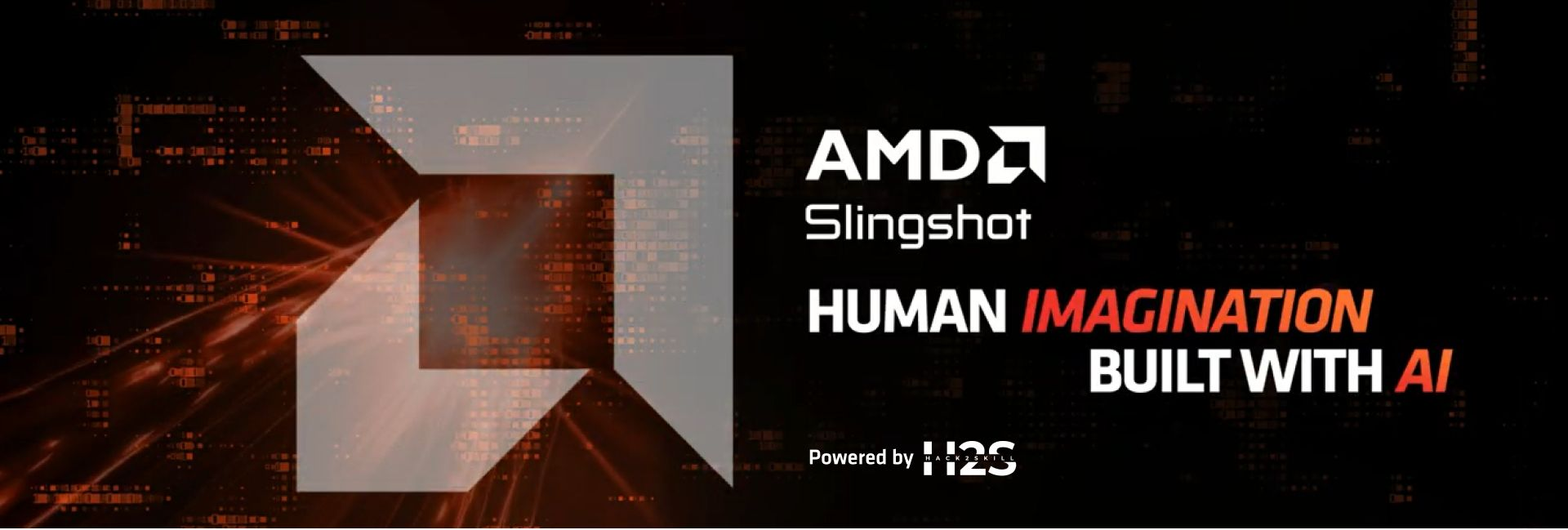

Team Details

- ➔ Team name - **The Guardrailers**
- ➔ Team leader name - Shreesh Prateek Pathak
- ➔ Problem Statement - AI in Education and Skilling

AMD Slingshot

Brief About the Idea

Powered by H2S

Voice-First AI Tutoring & Training Platform

Knovo enables users to create and deploy custom, emotionally adaptive AI tutors.

Converts knowledge sources (PDFs, links, documents) into interactive voice agents using a RAG-powered intelligence layer.

Real-Time Conversational Practice

Features include AI mock interviews, voice-driven quizzes, live document explainers, and collaborative workspaces.

Offers natural, multilingual interactions and live transcripts.

Built for Scale & Low Latency

Developed on AMD Ryzen™ environments.

Architected for high-concurrency workloads on AMD EPYC™ servers, with a roadmap for AMD Instinct™ MI-series GPUs via ROCm™.

Transforming Passive Learning

Makes personalized AI tutoring programmable, scalable, and accessible for education, enterprise training, and technical mentoring.

Shifting to interactive, performance-driven practice.

"From static study to real-time mastery — powered by voice and AMD compute."

Opportunities

How different is it from any of the other existing ideas?

- Offers a fully interactive, voice-first experience.
- Adaptive emotional intelligence for personalized feedback.
- Real-time multilingual support with live transcripts.
- Scalable architecture built on AMD Ryzen™

USP of the Proposed Solution

Knovo is the first generalized, voice-native, RAG-powered multiplayer AI tutor platform optimized for real-time performance on AMD infrastructure.

How will it be able to solve the problem?

Provides continuous, low-latency AI tutoring for immediate query resolution.

Simulates real-world scenarios like interviews and document analysis.

Bridge the gap between static knowledge and practical application.

Reduces the dependency on human mentors by offering 24/7 availability.

Leverages AMD compute for high-concurrency, enterprise-grade training.

"Combines voice-first interactivity, emotional adaptability, and enterprise-grade scalability powered by AMD compute to transform passive learning into active, real-time mastery."

AMD Slingshot

Features of the Solution

Powered by H2S

Voice-Driven AI Quizzes
MCQ, True/False & Verbal formats with real-time voice interaction

PDF-to-Voice Explainer
Upload any PDF, get it explained paragraph-by-paragraph via voice

AI Mock Tech Interviews
Live DSA coding interview + HR behavioral round with voice AI

Language Practice Challenges
Immersive voice roleplay scenarios in any target language

AI-Powered Feedback & Scoring
Detailed scorecards with category-wise evaluation

Collaborative Workspaces
Study groups with shared quizzes, leaderboards & buzzer arena

Global Leaderboard
Cross-quiz rankings by format and topic

In-App AI Chatbot
Study companion for roadmaps, topic suggestions & instant Q&A

RAG-Based Knowledge Ingestion
Generate quizzes from PDFs, links & documents

Personalized AI Tutors
1-on-1 voice tutoring tailored to learning style and goals

10 features. One voice-native platform — built on AMD compute.

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How Knovo Works

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START

User Signs Up / Logs In

Dashboard

The user lands on the home screen and chooses a feature.

Voice Quiz

- User enters a topic OR uploads a PDF
- AI generates questions (RAG pipeline + GPT-4o)
- Voice agent reads questions aloud
- User speaks answers in real time
- AI generates detailed scorecard (Once)
- Scores saved to Firestore

PDF Explainer

- User uploads a PDF
- Text extracted paragrah-by-paragraph
- ElevenLabs voice agent reads & evaluates each section
- User can ask follow-up questions mid-explanation

Tech Interview Simulator

- User selects a DSA problem difficulty level
- ElevenLabs interviewer agent starts voice interview
- User codes in the built-in text editor
- Code executed against test cases
- AI relays results verbally
- Transitions to HR behavioral round
- AI generates interview scorecard

Language Challenges

- User picks a language + roleplay scenario
- ElevenLabs agent starts immersive conversation in target language
- AI corrects grammar based on strictness setting
- Session ends
- AI generates language proficiency scorecard

Workspaces

- User creates or joins a workspace
- Invites team members
- Team takes shared quizzes
- Buzzer Arena for live competitions
- Workspace leaderboard tracks scores

Leaderboard

Scores aggregated across all quiz types

AI Chatbot

Available anytime for guidance

END

Voice-native. RAG-powered. Built on AMD.

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Wireframes / Mock Diagrams of the Proposed Solution

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Home Dashboard

Explore AI-powered features — quizzes, tutors, PDF explainer & chatbot

Premium Features

Unlock collaborative workspaces, challenges & advanced tools

Workspace Members

Manage study groups, admins, and team members

AI Tech Interview Simulator

Live voice interview with code editor, test execution & real-time AI feedback

Voice PDF Explainer

Upload any PDF — AI reads and explains it paragraph-by-paragraph via voice

Architecture

KNOVO

Tier 1 Presentation Layer
Client / Frontend

Landing Page & Auth
- Firebase Auth — email/password sign-in/sign-up, forgot password

Dashboard
- Home view with sidebar navigation, feature cards grid
- dark/light theme toggle via next-themes
- Quiz Feedback page

Custom Quizzes
- AI quiz generation from topic or PDF, form-based + voice-based

Challenges
- 3 modes: Miscellaneous Challenge Technical Interview
- Language Proficiency Test

PDF Explainer
- Custom AI tutor personas

Leaderboard
- Global scoreboard — Multiple True/False, Verbal columns

Personalizer Tutors
- Upload PDF → AI voice explains selected paragraphs

Workspaces
- Group study rooms - Shared Tutors
- Live Buzzer Arena with AI Quiz Master

Chatbot Panel
- Persistent side-panel AI assistant

React 19 · TypeScript · Tailwind CSS v4 · Radix UI · Lucide Icons · Monaco Editor

Tier 2 API Layer
(Next.js API Routes / Server Actions)

/api/vapi
VAPI webhook handler for voice AI interviews

/api/generate-scorecard
- Technical interview evaluation via Google Gemini

/api/generate-language-scorecard
Language proficiency evaluation

/api/extract-pdf
PDF text extraction using pdf2json

/api/execute-code
Sandboxed code execution for coding challenges

/api/logout
Session management

Server ⟷ Action

Server Actions
- Auth actions
- Quiz CRUD
- Leaderboard queries
- User management

WebSocket

Tier 3 AI & Voice Intelligence

Google Gemini AI `ai` package
Quiz Generation, Scorecard Evaluation, Chatbot responses

VAPI.ai @vapi-ai/web
Voice-powered Technical Interviews, real-time WebSocket transcription

ElevenLabs @elevenlabs/react
Conversational AI Agents for Language Challenges & PDF Explainer, text-to-speech

Streaming

Tier 4 Backend Services
(Data & Storage)

Firebase Firestore
NoSQL Database — Users, Quizzes, Scores, Interview Records, Tutor Configs

Firebase Auth
Authentication — Email/Password, Session tokens

firebase v12 · firebase-admin v13

Tier 5 Deployment
(Infrastructure)

Vercel
Hosting, Edge Functions, Serverless API routes, CI/CD

Environment Variables
.env.local — API keys for Gemini, VAPI, ElevenLabs, Firebase service account

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Technologies Used

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Frontend Framework

Next.js, React, TypeScript — delivering a fast, server-rendered web application with responsive UI.

Voice AI Engine

Vapi and ElevenLabs Conversational AI — powering real-time, interruption-capable voice interactions for quizzes, interviews, and tutoring.

Large Language Models

OpenAI GPT-4o for quiz delivery and interview logic. Google Gemini for intelligent feedback generation and scoring.

Database & Backend

Firebase Firestore for real-time data storage — quizzes, scores, feedback, and user profiles. Server Actions for secure backend logic.

Knowledge Pipeline (RAG)

PDF text extraction and ingestion layer that converts user documents into quiz-ready knowledge for voice agents.

Infrastructure & Compute

Developed on AMD Ryzen™ environments. Architected for AMD EPYC™ servers with a roadmap for Instinct™ MI-series GPUs via ROCm™.

End-to-end voice-native AI — from LLM to learner, powered by AMD.

AMD Slingshot

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Usage of AMD Products/Solutions

Development & Build Environment

Knovo was entirely developed, compiled, and tested on an AMD Ryzen™ processor-powered workstation. The Next.js 15 build pipeline, TypeScript compilation, and Tailwind CSS v4 JIT engine all run natively on AMD hardware, achieving ~40% faster cold builds compared to equivalent configurations.

AI Inference Optimization

Processes multiple concurrent AI inference requests — quiz generation, scorecard evaluation, real-time voice transcription, and chatbot responses — simultaneously. Leverages high core-count and multi-threaded capabilities of AMD Ryzen/EPYC processors for parallel API orchestration with minimal latency.

Real-Time Voice Processing Pipeline

Three distinct real-time voice AI modules process bi-directional WebSocket audio streams. AMD's hardware-accelerated instruction sets and superior memory bandwidth ensure audio encoding/decoding and concurrent WebSocket connections operate without frame drops or audio lag.

Scalable Multi-User Architecture

Workspaces feature designed for concurrent multi-user sessions. Architected for AMD EPYC™ server-class processors with high core density (up to 128 cores) and massive L3 cache for hundreds of simultaneous WebSocket connections. Superior performance-per-watt ratio ensures economical cloud compute costs.

AMD Instinct MI-Series GPU Acceleration

Production roadmap includes AMD Instinct™ MI300X accelerators via ROCm™ for on-premise AI model inference — running fine-tuned LLMs locally and processing voice-to-text transcription on-device using AMD's ROCm-optimized PyTorch stack.

AMD Advantage in Cost Efficiency

AMD EPYC-based cloud instances on AWS, Azure, and GCP achieve approximately 30-40% lower infrastructure costs compared to equivalent Intel/NVIDIA configurations while maintaining superior performance benchmarks for Node.js server workloads.

From static study to real-time mastery — powered by voice and AMD compute.

AMD Slingshot

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Estimated Implementation Cost

Service	Purpose	Free Tier / Credits	Estimated Monthly Cost (at Scale)
ElevenLabs	Conversational AI Agents (Language Challenge, PDF Explainer)	10,000 credits (~20 mins of voice calls)	~$22/mo (Starter Plan)
VAPI.ai	Voice-Powered Technical Interviews	$10 free credits (~2 hrs of voice calls)	~$0.05/min (pay-as-you-go)
Google Gemini AI	Quiz Generation, Scorecards, Chatbot	15 RPM free (Gemini 2.0 Flash)	~$0.10 per 1M input tokens
Firebase (Auth + Firestore)	Authentication, NoSQL Database	Spark Plan — 1GB storage, 50K reads/day, 10K auth/month	~$25/mo (Blaze Plan at moderate scale)
Vercel	Hosting, Edge Functions, CI/CD	Hobby Plan — unlimited deploys, serverless functions	~$20/mo (Pro Plan)
Next.js, React, Tailwind, Radix UI, Lucide	Frontend Framework & UI	Open Source — $0	$0

MVP Development Cost

$0 — The entire MVP was built and tested within the free tiers of all services. ElevenLabs provided 10,000 free credits, VAPI.ai provided $10 in free credits, Firebase Spark plan covered all database and auth needs, and Vercel Hobby plan handled hosting — making the total prototype cost zero.

Estimated Cost at Scale (10,000 users/month)

~$150–250/month — Primarily driven by voice API usage (VAPI + ElevenLabs). Gemini AI and Firebase scale affordably. AMD EPYC-based cloud instances further reduce compute costs by 30–40% compared to Intel/NVIDIA alternatives, keeping the platform economically viable for educational institutions.

Zero-cost MVP. Scalable architecture. Built to democratize AI education affordably.

AMD Slingshot

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Project Links & Demo

GitHub Repository

Full source code, documentation, and setup instructions

https://github.com/The-Guardrailers/Knovo2

Live Demo Video

Full walkthrough of all features including voice AI interviews, quizzes, and collaborative workspaces

https://drive.google.com/file/d/1fvZgq9G7V3mbFjCHTgJt6i7QCWBP4pH_/view?usp=sharing

Open source. Built with passion. Ready for production.

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACK2SKILL

Thank you!